Linear Regression:

given a certain amount of data, Predict the rest

$$y_i = \vec{m} x_i + b \Rightarrow y = X\vec{m} + b$$

$$\begin{bmatrix} x_{11} & \cdots & x_{1n} \\ \vdots & \ddots & \vdots \\ x_{m1} & \cdots & x_{mn} \end{bmatrix} \begin{bmatrix} m_1 \\ \vdots \\ m_n \end{bmatrix} + b = \hat{y}$$

* Z-Score Standardization is needed so we get better predictions from our regression. it compresses each value in X in a range between 0 and 1 which makes our weights have more of an effect on some properties that have lower original values.

* the formula for Z-Score standardization:

$$z = \frac{\overset{\text{Score}}{x} - \overset{\text{mean}}{\mu}}{\underset{\text{Standard deviation}}{\sigma}}$$

* Before doing anything we apply Z-Score to our inputs

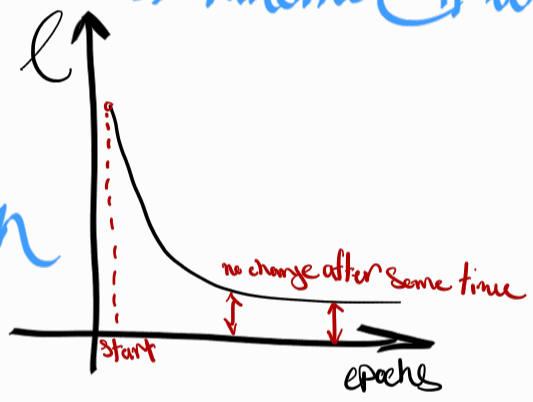
* Supervised learning means we can compare $\hat{y}$ to the original y. we call this the loss function

Loss function: (MSE)

$$J(\vec{m}, b) = \frac{1}{n} \cdot \left(\vec{y} - (X\vec{m} + b)\right)^2$$

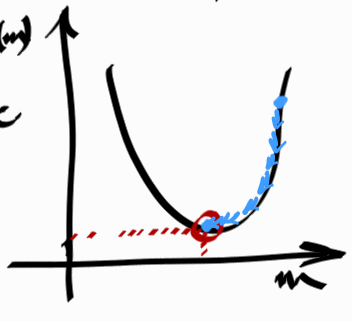
* if you plot the changes to the loss function it will look something like this:

after a certain time we will reach the minimum



* to minimize this function (to reduce the error of our model) we use gradient descent. we basically find the steepest slope and go the opposite way.

→ for a simple visualization imagine if our loss function had only one input "m". the plot of the function would look like a parabola and we want to reach the minimum of this parabola and by finding which way the function increases and going the opposite way we gradually make our way to the minimum



* Calculate the gradient:

$$\nabla_m J = \frac{\partial J}{\partial m} = -\frac{2}{n}\left(X^T\left(\vec{y} - \left(X\vec{m} - b\right)\right)\right)$$

$$\nabla_b J = \frac{\partial J}{\partial m} = -\frac{2}{n}\left(\vec{y} - \left(X\vec{m} - b\right)\right)$$

* now we gradually move the opposite way of our gradient until we reach the minimum the length of each step is determined by the learning rate (α)

→ Update the parameters:

$$\vec{m}_{new} = m - \left(\alpha \cdot \nabla_m J\right), \; b_{new} = b - \left(\alpha \cdot \nabla_b J\right)$$

thats basically it! easy right?